

Enhanced Online Continuous Brain-Control by Deep Learning-Based EEG Decoding

Jiaheng Wang¹, Lin Yao¹, and Yueming Wang¹

Abstract—Objective: A growing amount of deep learning models for motor imagery (MI) decoding from electroencephalogram (EEG) have demonstrated their superiority over traditional machine learning approaches in offline dataset analysis. However, current online MI-based brain-computer interfaces (BCIs) still predominantly adopt machine learning decoders while falling short of high BCI performance. Yet, the generalization and advantages of deep learning-based EEG decoding in realistic BCI systems remain far unclear. Methods: We conduct a randomized and cross-session online MI-BCI study on 2D center-out tasks in 15 BCI-naïve subjects. A newly proposed deep learning model named interactive frequency convolutional neural network (IFNet) is leveraged and rigorously compared with the prevailing benchmark namely filter-bank common spatial pattern (FBCSP) for online MI decoding. Results: Through extensive online analysis, the deep learning decoder consistently outperforms the clas-

sical counterpart across various performance metrics. In particular, IFNet significantly improves the average online task accuracy by 20% and 27% in two sessions compared with FBCSP, respectively. Moreover, a significant cross-session training effect is observed by the IFNet model ($P = 0.017$) while not for the controlled method ($P = 0.337$). Further offline evaluations also demonstrate the superior performance of IFNet over state-of-the-art deep learning models. Moreover, we present unique behavioral and neurophysiological insights underlying online brain-machine interaction. Conclusion: We present one of the first studies about online MI-BCIs using deep learning, achieving substantially enhanced online performance for continuous BCI control. Significance: This study suggests the good utility of deep learning in MI-BCIs and has implications for clinical applications such as stroke rehabilitation.

Index Terms—Brain-computer interface, motor imagery, deep learning, machine learning, online continuous control, EEG.

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Code: <https://github.com/Jiaheng-Wang/MI-BCI>

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I. INTRODUCTION

BRAIN-COMPUTER interfaces (BCIs) offer unique opportunities for patients suffering from various neurological disorders to restore daily life abilities or make a recovery after adequate neurofeedback therapy [1], [2], [3]. Among varieties of BCI methodologies, motor imagery (MI) which is one of the endogenous BCI paradigms has received intensive interest for a long time [4], [5], [6]. Using electroencephalography (EEG) as a non-invasive neuroimaging technique, it is observed that the amplitudes of sensorimotor rhythms (SMRs) can be voluntarily modulated by imagining movements [7], [8], which is referred to as event-related desynchronization (ERD) and event-related synchronization (ERS). To date, MI-BCIs have shown promise in various applications including cursor control [9], [10], wheelchair navigation [11], robotic arm control [12], [13], [14], video gaming [15], [16], post-stroke rehabilitation [17], [18], etc.

Despite great potential provided by MI-BCIs, MI decoding from EEG is a notoriously tricky problem [19], leading to low efficiency and restricted design choices in practical tasks and clinical applications. To this end, machine learning has long been introduced into the field of BCIs [20]. Based on the known neurophysiological signatures (ERD and ERS) [7], common spatial pattern (CSP) [21] and its variants such as filter-bank common spatial pattern (FBCSP) [22], have made pioneering efforts for the primary development of MI-BCIs [10]. More recently, deep learning has opened a new door

to further improve the potential of MI-BCIs by powerful neural network architectures [23], [24], [25]. For the MI decoding problem, representative CNNs such as Deep ConvNet, Shallow ConvNet [26], EEGNet [27], FBCNet [28], and IFNet [29] have shown their superiority against traditional benchmarks. While deep learning methods have been extensively investigated in offline settings, only a few studies have applied deep learning-based decoding in online practice. Moreover, the superior performance of deep learning methods has yet to be rigorously validated online.

Online MI-BCIs have so far been substantially investigated with traditional machine learning algorithms. One well-known MI-BCI named BBBCI is proposed by Blankertz et al [10]. They design a 1D cursor control task and utilize CSP with LDA for the two-class MI decoding. Further, McFarland et al [30] report that three-dimensional cursor control using EEG can be realized through a series of SMRs training sessions. They simply use the magnitude of power spectral density to perform continuous cursor control. Another remarkable application of MI-BCI is about robotic arm control for the 2D continuous pursuit task [13]. Edelman et al leverage electrical source imaging (ESI) to extract source-level band power and then a stepwise linear regression is used to select distinct features. Other studies using traditional machine learning algorithms include 3D quadcopter control [31], arm exoskeleton control [32], racing game [33], virtual hands control [34], to name a few. Typically, all of them make use of hand-crafted MI features while some also involve intensive subject training to meet expected performance. Nevertheless, classical feature engineering still falls short of high-performance MI decoding. As reported in [35], BCI control can not be achieved by at least 20% subjects, which is called the “BCI illiteracy” problem in traditional machine learning-based MI-BCIs.

Not until recently, a few studies have attempted to develop online MI-BCIs using deep learning-based EEG decoding [32], [34], [36], [37], [38]. Using data-driven end-to-end network architectures, they preliminarily demonstrate the feasibility and effectiveness of deep learning-based BCI control. Nevertheless, most of them simply employed a single deep learning method in their online implementations to verify the utility of their proposed MI-BCIs. Only one study [38] has compared different EEG decoding methods (one classical method using power spectral density (PSD) features, two deep learning methods) during 2D continuous pursuit. Notably, they found deep learning-based BCI control does not outperform the baseline counterpart at the first few sessions. Another study [39] surveyed existing EEG-based BCI frameworks that reported online classification performance. Similarly, they concluded that traditional methods are not surpassed by current deep-learning methods in online situations.

In fact, whether deep learning-based decoding methods designed in offline evaluation still keep advantages over traditional machine learning algorithms in online BCI control remains far unclear [39], [40], [41]. As opposed to offline decoding, online BCI control involves multiple aspects including paradigms, subjects, and models. (1) Online MI-BCIs can be applied to various applications such as cursor control, robotic arm control, wheelchair control, and so on.

In contrast, decoding models are often calibrated under simple cued MI tasks. Due to the non-stationary nature of brain activities, it is commonly observed that the performance of MI decoding algorithms degrades significantly across modes and sessions [42]. (2) When it comes to online BCI control, subjects continuously receive feedback information and in turn adjust their mental activities to accommodate to BCI systems, which can refer to a notion that “BCI is a skill to be learned” [43]. Hence, subject learning also plays an important role in online BCI control [44], which is intractable to be simulated and evaluated in offline analysis. (3) While classical MI decoding methods are derived from well-recognized neurophysiological priors, deep learning models are opaque and vulnerable to unexpected artifacts. As regards online feedback control, subjects might be more easily adapted to traditional decoding models as they are more neurophysiologically sound. In addition to these three factors, other factors such as device variations and user fatigue can also contribute to inconsistent data distribution between offline calibration and online control. Given the above serious concerns, there is an urgent need to conduct rigorous comparisons between traditional machine learning and newly developed deep learning methods in online MI-BCIs.

In this paper, we address the concerns about the generalization and advantages of deep learning in online BCI control from EEG. We leverage an improved version of our recently proposed deep learning model namely IFNet for online BCI decoding. Specifically, a log-power layer is adopted instead of a linear average one for temporal pooling. Moreover, a CutMix data augmentation strategy is introduced to further regularize network training. For the primary online experiment, our deep learning-based MI-BCI improves task accuracies by up to 27% when compared with the prevailing machine learning decoder in a group of BCI-naïve subjects. For the post-hoc offline evaluation, IFNet yields up to 7% higher accuracies against current state-of-the-art deep learning models. The main contributions of this work are as follows:

- We leverage deep learning-based EEG decoding to enhance online BCI control and systematically conduct both online multi-session performance comparisons with prevailing machine learning methods and offline evaluation with state-of-the-art deep learning models.
- Our deep learning-based MI-BCI system holds strong promise in continuous control tasks for BCI novices even in the initial session. It achieves significantly improved online performance as well as better training effect compared with existing benchmarks across sessions.
- We present comprehensive behavioral and neurophysiological insights regarding interaction effect between deep learning models and subjects underlying online feedback control.

II. RELATED WORK

In the field of MI-BCIs, a vast number of studies have been conducted on MI decoding based on EEG signals. A detailed review of MI decoding using machine learning algorithms can be found in [20]. Besides, a detailed review of deep learning in EEG is summarized in [25]. Combined with transfer learning,

TABLE I
ONLINE STUDIES OF MI-BCIs

Study	Task	MI types	Subjects	Method	Performance	Training
[9]	1D center-out	left/right-hand	5 healthy / 2 paralyzed	PSD [†]	66% acc. (4 targets)	✓
[45]	1D center-out	left/right-hand, both feet	13 healthy	CSP + LDA	81% acc. (2 targets)	×
[11]	wheelchair control	left-hand, rest, words association	2 healthy	PSD + Gaussian classifier	/	✓
[46]	1D center-out	left/right-hand, both feet	11 healthy	adaptive CSP + LDA	88% acc. (2 targets)	✓
[31]	3D quadcopter control	left/right-hand, both hands, rest	5 healthy	PSD [†]	/	✓
[17]	arm exoskeleton control	affected hand, rest	11 stroke	FBCSP + Naive Bayes	/	×
[12]	2D robotic arm grasp	left/right-hand, both hands, rest	13 healthy	PSD [†]	80% PVC (4 targets)	✓
[33]	racing game	both hands, both feet	2 paralyzed	PSD + Gaussian classifier	/	✓
[47]	FES feedback	left/right-hand	5 stroke	CSP + LDA	87% acc. (2 targets)	✓
[18]	FES feedback	affected hand, rest	13 stroke	PSD + Gaussian classifier	80% acc. (1 targets)	×
[13]	2D continuous pursuit	left/right-hand, both hands, rest	11 healthy	electrical source imaging	13% MSE	✓
[37]	multi-directional robotic arm reaching	arm reaching in multi-directions	15 healthy	CNN-LSTM	60% acc. (4 targets)	×
[48]	arm exoskeleton control	hand movement in left/right direction	16 stroke	PLV + Naive Bayes	69% acc. (2 targets)	×
[49]	virtual hands and FES feedback	left/right-hand	51 stroke	CSP + LDA	/	×
[36]	simple visual feedback	power/precision grasp	6 healthy	CNN-LSTM	65% acc. (4 targets)	×
[50]	VR and FES feedback	left/right-hand	20 healthy	CSP + SVM	87% acc. (2 targets)	×
[51]	virtual hands and FES feedback	affected hand, rest	17 stroke	CSP + SVM	73% acc. (1 targets)	×
[32]	arm exoskeleton and VR control	left/right-hand	20 healthy / 2 stroke	CNN with SE module	85% acc. (2 targets)	×
[52]	simple visual feedback	left/right-hand	14 healthy	weighted CSP + SVM	75% acc. (2 targets)	✓
[34]	virtual hands feedback	right-hand, rest	46 healthy	EEGNet	78% acc. (2 targets)	×
[53]	simple visual feedback	left/right-foot, rest	10 healthy	RFB-CSP + SVM	48% acc. (3 targets)	×
[38]	2D continuous pursuit	left/right-hand, both hands, rest	14 healthy	PSD [†] , EEGNet , PointNet	16% / 13% / 14% MSE	✓
[54]	simple visual feedback	left/right-hand	10 healthy	weighted CSP + LDA	67% acc. (2 targets)	✓
Our	2D center-out	left/right-hand, both feet, tongue	15 healthy	FBCSP+SVM, IFNet	49% / 59% acc. (4 targets)	×

FES: functional electrical stimulation. PSD: power spectral density. CSP: common spatial pattern. FBCSP: filter-bank CSP. LDA: linear discriminant analysis. SVM: support vector machine. CNN: convolutional neural network. LSTM: long short-term memory. SE: squeeze-and-excitation. PVC: percent valid correct. †: PSD features are extracted by autoregressive models using the BCI2000 software. Bold texts stand for deep learning-based methods.

deep learning has shown great improvements in both subject-dependent and subject-independent EEG decoding [55], [56], [57]. Despite significant advances in both machine learning and deep learning research on EEG decoding, existing studies methodologically build computational models and make comparisons by offline dataset analysis. While offline modeling is essential and efficient for constructing and optimizing data-driven BCI decoders, many factors can contribute to changes in the statistical characteristics of the data between offline BCI calibration and online BCI control [41], [58] (e.g. BCI applications, subject adaptation, and device variations). Hence, it is of vital importance to test the performance and user-friendliness of the whole BCI system in realistic online applications.

A summary of online studies of MI-BCIs is presented in Table I. As can be seen, these studies have shown the feasibility and effectiveness of MI-BCIs in diverse applications including cursor control, arm exoskeleton control, robotic arm control, etc. The majority of studies adopt left/right-hand MI to

perform BCI operations, while some also involve MI types of both hands, both feet, tongue, and rest. On top of healthy populations, MI-BCI is mostly applied to stroke patients for stroke rehabilitation of motor functions, demonstrating its profound clinical implications. However, with regard to MI decoding methods, existing MI-BCIs still heavily rely on traditional machine learning decoders whilst deep learning-based EEG decoders have not yet been extensively explored in online applications. More importantly, except for the study [38], all studies only employ one kind of decoding method (either machine learning or deep learning) for online BCI control. Therefore, it remains a critical question to what extent deep learning methods advance in real online BCI manipulation. The only study [38] encompassing two types of decoders has investigated MI-BCI in the 2D continuous pursuit task. On the one hand, they show the benefits of using deep learning models at the end of seven sessions, but the strength does not hold at the first few sessions. On the other hand, while

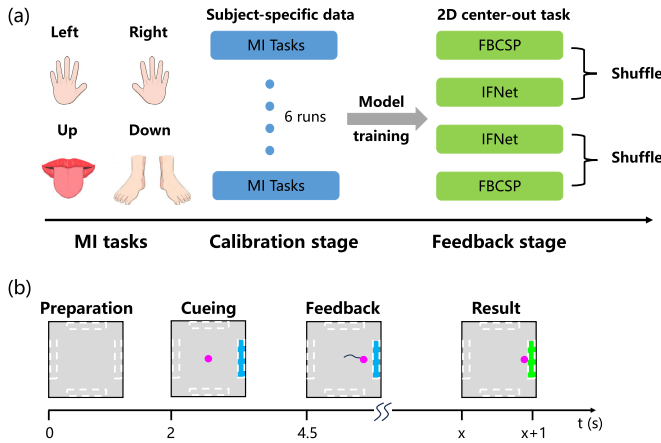


Fig. 1. Schematic illustration of MI-BCI experimental paradigm. (a) The main experimental design consisting of a calibration stage of six runs for model calibration and an online feedback stage of four runs performed by block-wise shuffled MI decoding algorithms. (b) Course of a feedback trial.

that study focuses on a complex pursuit task requiring BCI-experienced subjects, we believe investigating deep learning in a classical center-out task is also necessary to provide meaningful and straightforward instructions on nowadays MI-BCI applications, in particular, stroke rehabilitation [59]. It is noteworthy that some studies employ MI training strategies to further boost BCI performance. While BCI training might last for several sessions to weeks to take significant effect, it can be complementary to decoding algorithms, especially for subjects suffering from low BCI efficiency (e.g. <70% two-class accuracy). Taken together, to promote the utility of deep learning in online BCIs instead of currently prevailing machine learning approaches, we conduct systematic and rigorous model comparisons under online BCI scenarios.

III. MATERIALS AND METHODS

A. Experimental Design

Fifteen healthy subjects (average age: 24 ± 5.1 , twelve males) are recruited and participate in this two-session study after providing written informed consent to a protocol approved by the Ethics Committee of Zhejiang University (No. 2022005). Ten of them are fully naive to BCIs and the others have no experience of online BCI control. Subjects undergo two sessions of experiments within two weeks in an average intersession interval of 7.4 ± 5.4 days. The experimental paradigm is the same for both sessions. The main experimental design for each session is shown in Fig. 1 (a).

Concretely, subjects are sitting in front of a 24-inch monitor with arms lying relaxed on armrests. EEG signals are acquired with a 62-channel g.USBamp amplifier (g.Tec medical engineering, Schiedelberg, Austria) in an extended 10-20 system sampled at 256 Hz and highpass filtered above 0.1 Hz. For each session, the experiment consists of two stages, a calibration stage followed by a feedback stage. During the first calibration stage, subjects are instructed to perform four types of MI tasks, including imagined movements of the left hand, right hand, tongue, and both feet. The instructed four MI tasks are intended to map left, right, up, and down movements in

the subsequent feedback phase, respectively. There are six runs each containing 40 trials of randomly presented MI tasks with 10 trials per class. Each trial consists of 2 s of preparation, followed by 4 s of a visually cued MI task, and ended with roughly 2 s of rest. The indicator of an MI task is presented by a red arrow pinpointing the corresponding direction.

After data collection and model training, subjects undergo four feedback runs performing 2D center-out tasks in which subjects attempt to move the cursor outwards from the center of the screen to one of the four bars placed at the border of the screen. Each run is broken into 40 trials with balanced targets presented in a random order. An illustration of one feedback trial is shown in Fig. 1 (b). Each trial begins with a 2-s preparation period, followed by a 2.5-s cueing period where the blue-colored target is informed and subjects start to perform the corresponding MI task. After that, MI feedback is provided while subjects continuously control the cursor to hit the target. The feedback period ends either upon a collision with a bar or after 10 s with no collision. At the end, a 1-s result period displays the green or red-colored bar depending on whether or not it is the correct target.

To mitigate time and learning effects which might bias and confuse comparisons in regard of two MI decoding algorithms, FBCSP and IFNet are randomly shuffled in each feedback block. Specifically, the four feedback runs are further divided into two blocks (two runs per block). In each block, FBCSP and IFNet are randomly applied to one of the two runs, respectively. Moreover, subjects are not aware of which algorithm they are experiencing. Individuals are instructed to try their best to perform desired MI tasks regardless of the feedback results. There are no adverse events or abnormal behaviors emerging for any subject.

B. Filter-Bank Common Spatial Pattern

For traditional machine learning algorithms, we adopt FBCSP [22] which is representative and benchmarking for MI decoding. FBCSP was adopted by the winning entry of the 2a dataset in the BCI competition IV and is most commonly used for comparison with deep learning models. We use the implementation available in the MOABB toolbox [60] with custom scripts. Furthermore, We perform probability estimates of the output of SVM and implement it using the *libsvm* package [61]. On the one hand, leveraging the probability output of SVM in the range of 0-1 could partly deal with unforeseen outliers since the maximum speed of the cursor is under control. On the other hand, it bridges the gap between the output range of linear models and deep learning models, allowing a uniform cursor control strategy. We further validate FBCSP with probabilistic SVM on the BCIC-IV-2a dataset. It yields an average accuracy of 69.4% which is comparable to the performance of the official FBCSP implementation (67.59%). This underlines that FBCSP with probabilistic SVM is also suitable for online operation.

C. Interactive Frequency CNN With CutMix

For deep learning models targeting MI decoding, a recently proposed lightweight CNN named IFNet is adopted for

the state-of-the-art performance across several public datasets [29]. Principally, IFNet is inspired by the concept of cross-frequency coupling (CFC) that represents the association of multiple-frequency neural oscillations [62]. Thereby, frequency interactions between low and high-frequency bands are explicitly constructed in IFNet, enhancing the representation of MI characteristics. Technically, the IFNet decoding model works as follows. EEG signals are first filtered into low (4-16 Hz) and high-frequency (16-40 Hz) bands, resulting in $X_l \in \mathbb{R}^{C \times T}$ and $X_h \in \mathbb{R}^{C \times T}$ respectively. Then, 1D spatial convolution $K_s \in \mathbb{R}^{C \times 1}$ followed by 1D temporal convolution $K_t \in \mathbb{R}^{1 \times w}$ is performed on each filtered band. Next, an element-wise addition operation is adopted to explicitly model the interplay between the two frequency bands, resulting in $U \in \mathbb{R}^{F \times T}$. Then, a subsequent temporal pooling layer downsamples the integrated features using average pooling, yielding $U \in \mathbb{R}^{F \times W}$. In the end, spectro-spatial-temporal features are fed into a fully connected layer with the softmax transformation for the final probability prediction. Concerning model size and training complexity, IFNet is lightweight which has approximately 10K trainable parameters and it takes roughly 3 min per training fold using an NVIDIA GeForce RTX 2080Ti graphics card. For a detailed illustration of IFNet, please refer to our previous work [29] and corresponding open-sourced code.

Based on the previous work, we further progress the temporal pooling block using a logarithmic power operator as what the CSP algorithm does instead of a linear average one. In particular, this is similar to the log-variance layer used in FBCNet [28]. Mathematically, they are defined by the second raw moment and second center moment followed by a logarithmic activation in IFNet and FBCNet, respectively. The new implementation of IFNet (IFNet V2) yields an average classification accuracy of 79.72% on the BCIC-IV-2a dataset [63]. Moreover, it improves the accuracy by 1% and 2% compared to IFNet V1 on the OpenBMI dataset [64] in the within-session and cross-session analysis, respectively.

To further enhance network training, a new data augmentation strategy named CutMix is proposed. As shown in Fig. 2, CutMix stands for the *cut-and-then-mix* operation which is simple yet effective. Consider two EEG trials $S_1 \in \mathbb{R}^{C \times T}$ and $S_2 \in \mathbb{R}^{C \times T}$ belonging to the same MI type, where C denotes EEG channels and T denotes sampling points. To perform CutMix, a cutting window is randomly generated with a range of $[t_1, t_2]$, where t_1 and t_2 represent the start and end time point of the cutting window. Then, the cutting signals from the two EEG trials are exchanged as follows:

$$\begin{aligned} S_1[1 : C, t_1 : t_2] &:= S_2[1 : C, t_1 : t_2] \\ S_2[1 : C, t_1 : t_2] &:= S_1[1 : C, t_1 : t_2] \end{aligned} \quad (1)$$

The new created EEG trials S'_1 and S'_2 will serve as the additional training samples for each mini-batch training. By doing this, we regularize the network to concentrate on task-specific stationary characteristics rather than trial-specific artifacts due to insufficient and noisy calibration data.

The network training is conducted on the calibration data. We adopt the same training procedure as that in the original

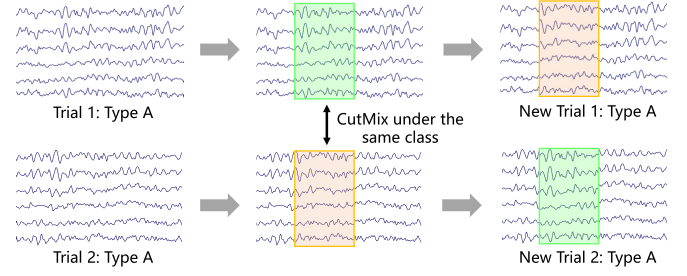


Fig. 2. An example illustration of CutMix (cut-and-then-mix) data augmentation.

paper. To verify the generalization of IFNet, all the hyperparameters are set in their default values. We conduct our experiments on an individual computer equipped with an Intel Core i9-10900k processor and an NVIDIA GeForce RTX 2080Ti graphics card. In terms of computational efficiency, it takes up to 3 min for one network training and roughly 5 ms per inference time, manifesting IFNet is applicable to online continuous operation.

D. Online BCI Control

For online BCI control, 35 electrodes covering the sensorimotor region of the scalp are used for online processing, including F5, F3, F1, FZ, F2, F4, F6, FC5, FC3, FC1, FCZ, FC2, FC4, FC6, C5, C3, C1, Cz, C2, C4, C6, CP5, CP3, CP1, CPZ, CP2, CP4, CP6, P5, P3, P1, PZ, P2, P4, P6. EEG signals in bad channels are visually identified and interpolated through spherical splines using the MNE python package [65]. During online feedback control, the most recent 2 s of EEG signals are extracted and fed into decoding models for the real-time continuous classification every 40 ms. The predicted MI type is then translated to the cursor movement, continuously updating the cursor position on a screen.

The translation strategy is as follows. First, the class C with the highest probability P is determined by the decoding model for each frame. Then, P is corrected by subtracting its minimum value of 0.25 to eliminate the origin offset. That is, the offset-corrected P value is in the range of $[0, 0.75]$ so that the origin state of 0 would exert no effect on cursor movement, which otherwise would have a value of 0.25. Next, V_x and V_y representing control intensities in horizontal and vertical directions are derived according to the decoded class C .

$$\begin{cases} V_x = -P, V_y = 0; & \text{if } C \text{ is left-hand MI} \\ V_x = P, V_y = 0; & \text{if } C \text{ is right-hand MI} \\ V_x = 0, V_y = P; & \text{if } C \text{ is tongue MI} \\ V_x = 0, V_y = -P; & \text{if } C \text{ is both feet MI} \end{cases} \quad (2)$$

Then, the movements of the cursor in each dimension D_x and D_y are obtained using a momentum-based updating strategy as shown below.

$$\begin{aligned} D_{xnew} &= \mu D_{xold} + \gamma(1 - \mu)V_x \\ D_{ynew} &= \mu D_{yold} + \gamma(1 - \mu)V_y \end{aligned} \quad (3)$$

where μ represents the momentum coefficient and γ represents a scalar gain. At the initial stage, D_x and D_y are set to 0,

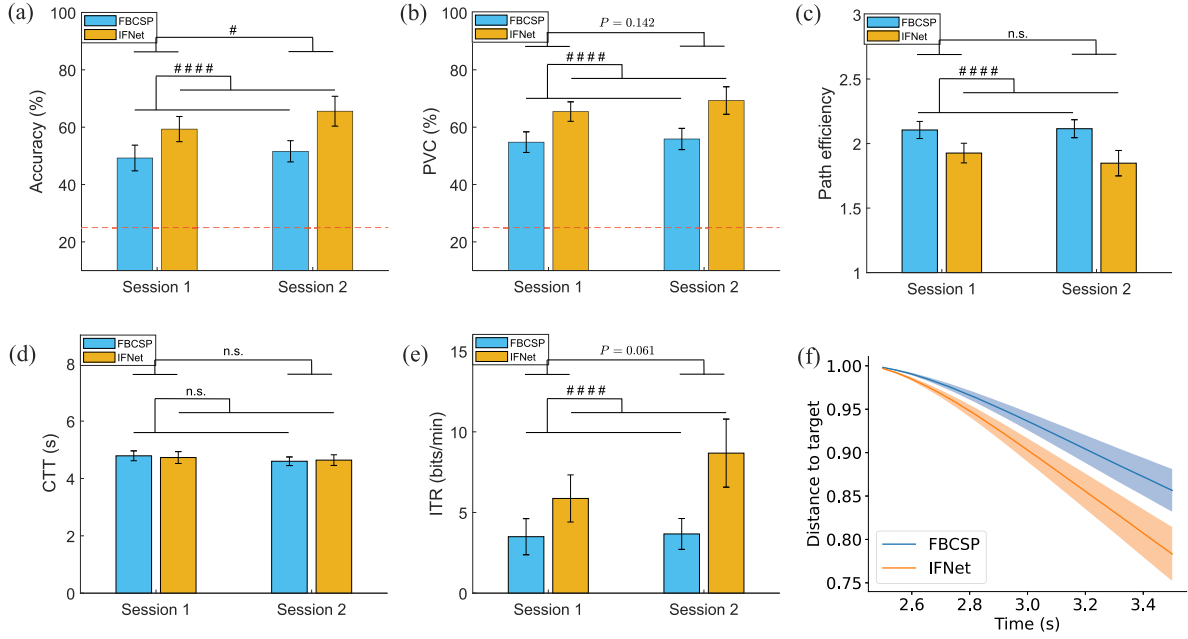


Fig. 3. Performance metrics in online 2D center-out tasks. (a, b, c, d, and e) Subject-averaged task accuracy, percent valid correct (PVC), path efficiency, correct trial time (CTT), and information transfer rate (ITR) for FBCSP and IFNet in two sessions, respectively. The red dotted line indicates the chance level. (f) Grand-averaged cursor distance to target over feedback time for the two models. Bars indicate mean + standard error of the mean. Statistical analysis uses two-way repeated-measures ANOVA with the main effects of model and session. # $P < 0.05$, #### $P < 0.0001$.

respectively. μ is set to 0.8. Since the maximum value of P is 0.75, we set γ to 25, meaning that it takes at least 1 s to successfully achieve one center-out task. In the sequel, the positions of the cursor L_x and L_y are updated relatively.

$$\begin{aligned} L_{xnew} &= L_{xold} + D_{xnew} \\ L_{ynew} &= L_{yold} + D_{ynew} \end{aligned} \quad (4)$$

E. Performance Metrics

Online performance metrics are evaluated and compared between FBCSP and IFNet in both sessions. Specifically, there are five performance metrics with respect to 2D center-out tasks in the feedback stage. (1) Accuracy is computed as the number of correct trials divided by the total number of trials. (2) Percent valid correct (PVC) [66] is calculated as the number of correct trials divided by the number of valid trials, where a final decision is made. (3) Path efficiency is used to quantify the goodness of cursor trajectories. It is measured as the total distance traveled en-route to the target normalized by the straight-line distance from the starting location to the target. (4) Correct trial time (CTT) is defined as the average time length of the correct trials. (5) Information transfer rate (ITR) [1] is utilized to take into account both speed and accuracy. Offline performance is measured by decoding accuracy on trial samples with fixed decoding windows (e.g., 0.5 s - 2.5 s), as is often the case in offline evaluation.

F. Statistical Analysis

For online performance metrics, two-way repeated-measures ANOVA is employed with two within-subject factors of session (2 levels: Session 1 and Session 2) and model (2 levels: FBCSP and IFNet) using the *rstatix* package (0.7.2). The

normality of metrics is visually checked by histograms and QQ plots. When the assumption of normality is clearly violated for a certain metric, then a box-cox transformation is applied to that metric before performing the ANOVA test. Significant findings are followed by independent post-hoc tests to aid inference. All post-hoc inferences are two-sided at a Bonferroni corrected alpha level of 0.05 (0.025 per side). As for pairwise comparisons in offline evaluation, a paired t-test with Bonferroni correction is used when the residual of the data is normally distributed, otherwise a Wilcoxon signed-rank test with Bonferroni correction is applied.

IV. RESULTS

A. Online Performance Analysis

Online performance metrics regarding 2D center-out tasks are presented in Fig. 3. As a whole, IFNet demonstrates superior performance in comparison with FBCSP across metrics and sessions. First of all, in terms of task accuracy, IFNet achieves 51.58% and 65.58% average accuracy in two sessions, respectively, improving those of FBCSP by 20% and 27%. A two-way repeated-measures ANOVA reveals the significant main effects of both session ($P = 0.011$) and model ($P < 0.0001$) on mean accuracy. Besides, there is no significant interaction between session and model ($P = 0.296$), indicating the improvement from session 1 to session 2 is not significantly different between FBCSP and IFNet. Since there are only two levels in both factors, pairwise comparisons are not performed. Another analogous metric namely PVC shows similar results except for an insignificant main effect of session ($P = 0.142$). Next, for the path efficiency metric, IFNet significantly outperforms FBCSP by nearly 10% and 14% ($P < 0.0001$) in both sessions, respectively. In addition,

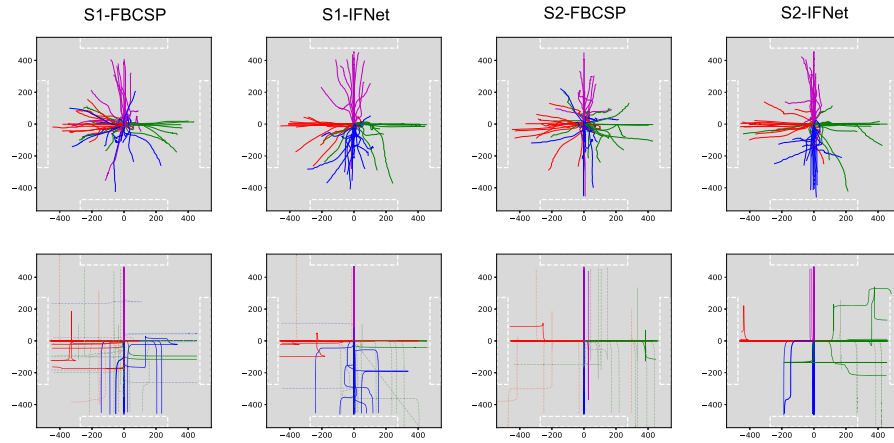


Fig. 4. Illustration of cursor trajectories during feedback control. The first row presents the trial-averaged cursor trajectory of each task for all subjects using different models. The second row presents the cursor trajectory of each trial for one representative subject, wherein the solid curves denote correct trials and the dashed curves denote failed trials. Red, green, pink, and blue curves denote center-out tasks toward left, right, top, and bottom targets, respectively. It is observed that IFNet responses more accurately and stably compared with FBCSP in general. Moreover, it obtains more apparent improvements over sessions.

TABLE II
OFFLINE FOUR-CLASS CLASSIFICATION ACCURACIES OF DIFFERENT MODELS IN CV AND HO ANALYSES

Training Set	Test Set	FBCSP-SVM	EEGNet	FBCNet	Conformer	TSFCNet	IFNet
S1-Calibration	CV	59.69	65.39	67.31	61.39	64.25	70.53
	S1-Feedback	49.29	55.97	55.08	52.54	54.24	59.97
	S2-Calibration	48.17	55.46	57.77	55.70	55.91	61.36
	S2-Feedback	41.54	49.78	48.95	49.08	49.33	54.73
S2-Calibration	CV	63.81	69.69	72.14	66.72	67.81	76.06
	S2-Feedback	52.71	60.50	60.43	57.93	57.11	63.68

CV denotes five-fold cross-validation analysis, and HO denotes hold-out analysis using an independent test set. IFNet significantly outperforms other methods in all analyses (all $P < 0.01$).

neither the session effect ($P = 0.474$) nor the interaction effect ($P = 0.273$) is found to be statistically significant. In contrast to the above metrics, almost identical CTT results are observed for the two models across sessions. Accordingly, no significant differences can be drawn (session: $P = 0.356$, model: $P = 0.525$, session $\times$ model: $P = 0.651$). It might be explained by the unified speed gain γ of cursor movement used throughout the whole experiment. While it allows model comparison in a uniform condition, subject-specific speed gain can be applied to further enhance BCI efficiency. The ITR metric simultaneously considering accuracy and speed shows a significant difference of average ITR between the two models ($P < 0.0001$). Besides, a near-significant main effect of session is observed ($P = 0.061$). In detail, IFNet improves average ITR over sessions by nearly 50% while FBCSP moderately improves it. The last panel shows the grand-averaged cursor distance to target over feedback time. As seen, IFNet yields better responsiveness to the intended target.

To provide more insights of trajectory dynamics performed by the two models, illustrations of cursor trajectories during feedback control are shown in Fig. 4. The upper panel shows the trial-averaged cursor trajectory for each task across subjects. In account of variations on the time lengths of trials, trials belonging to the same task (e.g., center-to-left) are first downsampled to the minimum trial length within this

group. As seen, IFNet yields more accurate and stable control trajectories compared with FBCSP. Also, more apparent cross-session improvements are attained by IFNet. The lower panel presents detailed trial curves traced by one representative subject. For this subject, a superior error-correcting ability (green and blue solid curves in S2-IFNet) is observed by IFNet while the subject has trouble rectifying initial errors by FBCSP (green dashed curves in S2-FBCSP).

B. Offline Model Comparisons

We further conduct offline evaluation involving diverse deep learning models. Specifically, four additional deep learning models namely EEGNet [27], FBCNet [28], EEG-Conformer [67], and TSFCNet [68] are investigated to evidence the advantages of deep learning models. The same training procedure and hyperparameter settings are applied following the previous works. A time window of 0.5 s - 2.5 s after the cue onset is used for offline model evaluation which is consistent with the 2-s-long decoding window used in online cursor control. Analyses are performed under the different combinations of the calibration and feedback stages of the two sessions. The average four-class and two-class decoding accuracies of different models are shown in Table II and Table III. We have the following observations: (1) All deep learning models outperform FBCSP+SVM by a large margin across all test configurations. (2) IFNet consistently

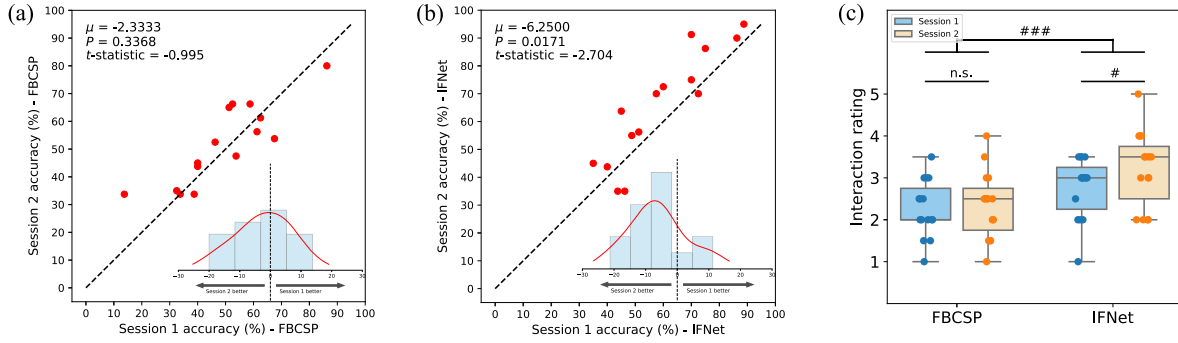


Fig. 5. (a) and (b) Comparative performance of online BCI control from session 1 to session 2. Red dots represent the accuracies of the subjects. (c) Subjects' ratings in terms of BCI experiences. $\#P < 0.05$, $###P < 0.001$. The results show a significant cross-session training effect and better user experiences of IFNet compared with FBCSP.

TABLE III
OFFLINE TWO-CLASS CLASSIFICATION ACCURACIES OF DIFFERENT MODELS IN CV AND HO ANALYSES

Training Set	Test Set	FBCSP-SVM	EEGNet	FBCNet	Conformer	TSFCNet	IFNet
S1-Calibration	CV	68.11	70.56	69.33	69.61	71.39	73.50
	S1-Feedback	62.08	67.12	64.07	65.13	65.52	68.83
	S2-Calibration	63.61	66.09	68.52	68.32	67.24	70.47
	S2-Feedback	58.75	63.98	64.53	64.30	65.03	67.95
S2-Calibration	CV	72.72	74.72	71.00	72.67	73.78	78.22
	S2-Feedback	66.83	69.42	69.08	67.13	69.47	73.20

CV denotes five-fold cross-validation analysis, and HO denotes hold-out analysis using an independent test set. IFNet significantly outperforms other methods in all analyses (all $P < 0.01$).

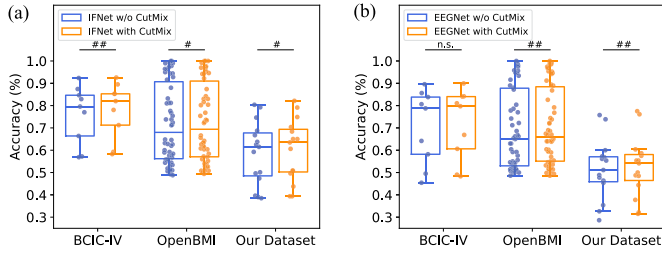


Fig. 6. Ablation studies on CutMix with IFNet and EEGNet across the three datasets. $\#P < 0.05$, $##P < 0.01$.

achieves the highest average accuracy compared with the others (all $P < 0.01$). (3) Both cross-session and cross-mode decoding result in a considerable decrease of accuracy when compared with the cross-validation analysis. (4) Offline cross-mode evaluation exhibits nearly the same performance as that in online assessment. For example, using S1-calibration as the training set, IFNet yields 59.97% four-class accuracy on the S1-feedback test set while the corresponding online accuracy reaches 59.33% as well. This supports that offline analysis could provide important references for online continuous control tasks. (5) Cross-session learning effect is consistently observed across models, while subjects tend to be more adaptive to deep learning methods. To sum up, deep learning models universally surpass traditional machine learning methods in both online and offline evaluations.

To verify the effectiveness of the proposed CutMix, we conduct ablation studies regarding CutMix on three MI datasets. Fig. 6 presents four-class classification accuracies on the BCIC-IV and our datasets as well as two-class classification

accuracies on the OpenBMI dataset. As demonstrated, both IFNet and EEGNet achieve significant improvements across different datasets when trained with the CutMix augmentation.

C. Cross-Session Training Effect

As 'BCI is a skill to be learned' [43], it is of great interest to investigate and compare the training effect of deep learning and machine learning on subject adaptation over sessions. Detailed comparisons of subjects' task accuracies from session 1 to session 2 are presented in Fig. 5 (a) and (b). As we can see, the improvements with IFNet are remarkable for the majority of subjects while the improvements with FBCSP are marginal. In particular, there are 80% subjects exhibiting improved task performance operating on IFNet-based MI-BCIs, while roughly 50% subjects gain improved BCI control using FBCSP as the decoder. The mean task accuracies improved by IFNet and FBCSP are 6.25% and 2.33%, respectively. Further pairwise t-tests suggest a significant cross-session improvement by IFNet ($P = 0.017$), whereas FBCSP does not ($P = 0.337$). We further present analysis from a subjective point of view. Specifically, at the end of each experimental session, subjects assess their BCI experiences with respect to each feedback run, measured by a score from one to five. (How do you feel about operating the BCI system at that run? One stands for very hard, and five stands for very good). The results are plotted in Fig. 5 (c). It can be seen that IFNet turns out to offer significantly better BCI experiences compared with FBCSP across both sessions. Besides, subjects gain significantly improved BCI experiences over sessions when operating on IFNet-based MI-BCIs whereas no obvious improvement is observed

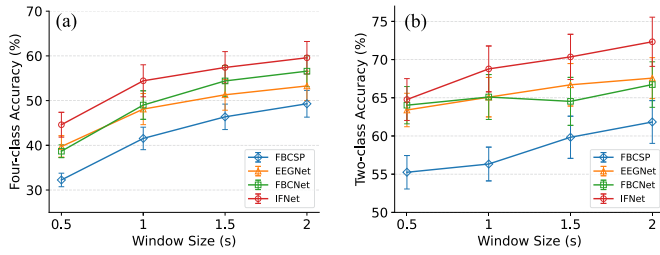


Fig. 7. The sensitivity of decoding models to window sizes by offline evaluation on four-class and two-class MI decoding, respectively. The error bars represent standard mean errors.

for the FBCSP counterpart. These empirical findings collectively suggest that deep learning-based MI-BCIs possess the remarkable ability to enhance BCI training and are more user-friendly when compared with conventional machine learning approaches.

D. Sensitivity Analysis on Decision Windows

For an online MI-BCI system, a short decision window implies a short latency in response to user intent and vice versa. On the one hand, lower latency allows more fine-grid and sophisticated BCI control. On the other hand, decoding accuracy is generally inversely proportional to the length of the decoding window. Therefore, it is of great concern to investigate the trade-off between accuracy and latency. In our experiment, a moderate 2-s-long window length is used to maintain stable and reliable decoders during 2D center-out tasks, since latency is not critical in this sense. However, this is not the case when it comes to more complicated 2D tasks such as continuous pursuit tasks, reach and grasp tasks, and so on. To this end, we conduct sensitivity analysis regarding decision window length using a similar procedure as in Section IV-B. Concretely, four decoding windows of 0.5 s - 1 s, 0.5 s - 1.5 s, 0.5 s - 2 s, and 0.5 s - 2.5 s after the cue onset are analyzed across different models with cross-mode evaluation.

Both four-class as well as two-class classification accuracies with different window sizes and models are investigated and presented in Fig. 7 (a) and (b), respectively. Here, all models show a noticeable sensitivity to window sizes, while deep learning models still maintain their advantages over FBCSP. Specifically, IFNet achieves the highest performance across different windows, followed by FBCNet. As for the four-class MI decoding, dramatic declines are observed between 1-s-long and 0.5-s-long windows for all models, indicating lack of distinguishable and robust information for EEG-based MI decoding in a very short latency. Contrarily, the accuracy moderately drops when reducing window length from 2 s to 1 s, in particular, 5% by IFNet. When it comes to the two-class MI classification, a nearly 70% average accuracy is achieved by IFNet under the window length of 1 s, which is assumed to be a threshold required for BCI applications. Consequently, a 1-s-long decision window may strike a balance between accuracy and latency and could be taken into account in future study.

TABLE IV

CLASSIFICATION ACCURACIES IN OFFLINE AND ONLINE ANALYSES

Model	Session	FBCSP runs	IFNet runs	Online acc.
IFNet	S1	59.97	59.97	59.33
	S2	62.98	64.38	65.58
FBCSP	S1	49.25	48.50	49.25
	S2	51.92	53.50	51.58

V. DISCUSSION

A. Transition From Offline to Online Analysis

As opposed to previous deep learning works that are dedicated to developing effective network architectures using standard, publicly available datasets [25], we conduct online BCI evaluation and validate the generalization and effectiveness of deep learning models. Online feedback control involves distinct user adaptation that interacts with decoding models, not just the machine side [69]. Yet, performance comparisons between offline and online evaluations remain to be investigated, particularly whether offline calibration results generalize well to online BCI applications. To this end, we first conduct offline performance evaluations on the two investigated decoding models. Specifically, the feedback stage is divided into homogeneous runs and heterogeneous runs with respect to the model tested. That is, for IFNet evaluation, feedback runs implemented with IFNet represent homogeneous runs while the others represent heterogeneous runs, and the same applies to FBCSP. In that case, model evaluation on heterogeneous runs stands for the common offline analysis while model evaluation on homogeneous runs can be regarded as online analysis regarding discrete control tasks. A time window of 0.5 s - 2.5 s is used to evaluate decoding accuracy under both conditions. Those results are further compared with the real online performance in continuous control tasks. The results are presented in Table IV. We find that, offline evaluation attains almost the same subject-averaged results as online experiments across both models and sessions. Further statistical analyses show no significant discrepancy between offline and online evaluations. Consequently, in a general sense, offline evaluation renders generalized estimations for online situations which are not biased by specific models implemented in the feedback stage. Nevertheless, the above interesting phenomena are observed in relatively simple 2D center-out tasks in which latency is not of particular importance. Further investigation is required in latency-critical settings.

B. Neurophysiological R^2 Patterns

Offline results show consistently remarkable deterioration of performance when translating CV analysis to cross-mode and cross-session analysis, indicating large inter-mode and inter-session variability. What's more, it is of interest to investigate how individuals modulate their brain patterns in response to different models. We attempt to explore performance variations and brain-machine interactions from the neurophysiological perspective. To this end, the coefficient

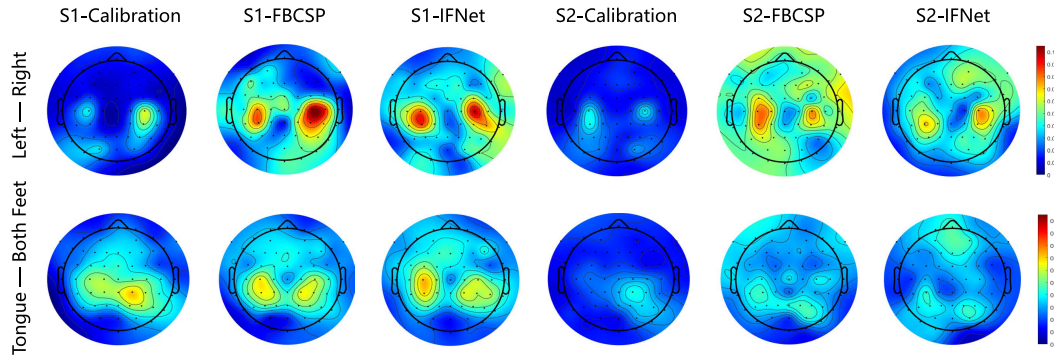


Fig. 8. Grand-averaged R^2 patterns within the upper alpha band (10-13 Hz) over a period of 0.5 s - 2.5 s after the cue onset across modes and sessions. The first row presents R^2 maps of left-hand vs. right-hand in the calibration and model-specific feedback stages in session 1 and session 2, respectively. The second row presents R^2 maps of tongue vs. both feet. On the one hand, heightened and more global brain discriminative patterns can be observed in feedback stages. On the other hand, R^2 patterns are less prominent in the second session. In addition, brain topographies are relatively consistent across the two models in both sessions. Color bars indicate R^2 values.

of determination R^2 is adopted reflecting how much of the variance in the distribution of all samples is explained by the class affiliation [70]. Briefly, it calculates a squared Pearson correlation coefficient between spectral power values and binary class labels at each sensor. Specifically, a time period of 0.5 s - 2.5 s after the cue onset and an upper alpha band of 10-13 Hz are selected in knowledge of neurophysiological priors of MI. We separate the four MI types into horizontal (left - right) and vertical (tongue - both feet) pairs, respectively.

Fig. 8 shows grand-averaged R^2 patterns across modes and sessions. On average, the most discriminative information is located in sensorimotor areas around the C3 and C4 electrodes, which is in accordance with well-known ERD/ERS signatures of MI. Importantly, evident inter-mode and inter-session variabilities are observed. As for inter-mode comparisons, a globally moderate improvement of discriminability from the calibration to feedback stage can be observed in both sessions. We contemplate that subjects manage to adapt themselves according to feedback results and therefore induce more distinguishable brain patterns. As for inter-session comparisons, the average R^2 maps tend to be less distinct in the second session. This is not intuitive since BCI performance shows consistent improvements over sessions. One plausible explanation is about inter-subject variability with respect to MI patterns. The advanced capability of deep learning models allows to capture subject-specific MI characteristics differing from the standard ones as well as alleviate subjects' effort to achieve prompted tasks. Once subjects learn to operate their own discriminative brain signals, group-level features might be less focal and prominent. In the view of inter-model comparisons, similar discriminative patterns are observed for the two models in both sessions, giving further confidence that the discrepancy of task performance is indeed attributed to the models themselves. Apart from the sensorimotor area, frontal, temporal, and occipital areas are also found to contain MI-discriminative features, especially in the S2-feedback stage. These observations additionally suggest that MI-BCI control makes use of a distributed brain network concurrently dealing

with motor imagery, attention, awareness of feedback, and so on [71].

C. Network Interpretability Analysis

To gain more insights into performance degradation from offline calibration to online decoding, we conduct network interpretability analysis on deep learning models, explaining performance variations in brain-machine interaction from the machine perspective. We adopt a gradient-based attribution method called Integrated Gradients¹ [72] to attribute predictions of a deep learning model to its input signals. As regards EEG decoding, it could tell us which sample points of an EEG trial are responsible for a certain label (MI type) being picked. The general formula of integrated gradients is shown below:

$$\text{IG}_i = (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha \quad (5)$$

where x_i and x'_i denote the i^{th} elements of the input and baseline, F denotes model output, α denotes the interpolation constant from 0 to 1. Subject-level MI attribution patterns are calculated as follows. First, integrated gradients are calculated on each EEG trial with its corresponding MI type, yielding $G \in \mathbb{R}^{C \times T}$ for each trial. Next, absolute attributions $|G|$ are averaged along the temporal dimension, resulting in $G' \in \mathbb{R}^C$. Then, attributions of trials with the same MI type are averaged, thus subject-level MI attribution patterns $\bar{G} \in \mathbb{R}^C$ are obtained. As the result, the larger the attribution value of a channel, the greater its contribution to the MI prediction. We compute attribution patterns in both the calibration and feedback stages. Of note, IFNet models used for attribution analysis are separately trained with respective data in the two stages.

Attribution patterns for the two representative subjects are shown in Fig. 9 (a) and (b). Subject A attains both a high calibration accuracy and online accuracy of 93% and 95%, respectively. While for Subject B, a high calibration accuracy of 86% but a low online accuracy of 56% is achieved. We have

¹<https://github.com/pytorch/captum>

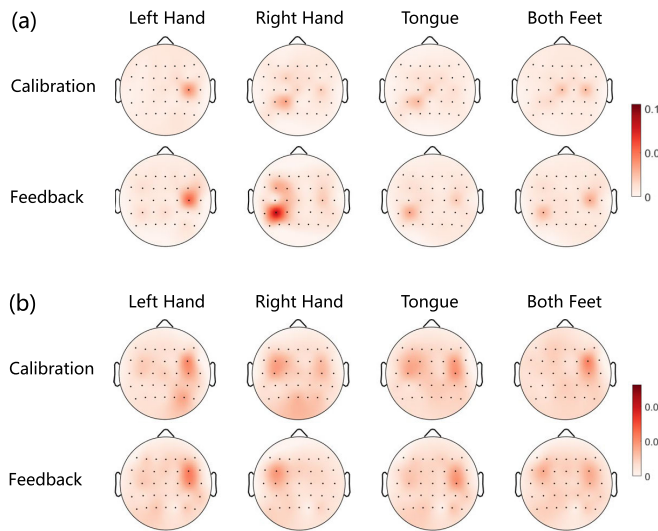


Fig. 9. MI attribution patterns by IFNet based on integrated gradients for the two representative subjects in both the calibration and feedback stages. (a) Subject A achieving both high calibration and online accuracies. (b) Subject B achieving high calibration yet low online accuracies. As seen, attribution patterns are closely associated with the well-recognized MI signatures. Besides, less clear and consistent model attributions are observed by Subject B, which explains his severe performance degradation from calibration to feedback control. Color bars indicate averaged absolute values of integrated gradients.

the following observations. (1) On the whole, channels with large attributions mainly lie in the sensorimotor area, which are closely associated with the well-recognized MI-activated brain regions. In particular, when performing left-hand MI tasks, the most prominent MI signatures are attributed to the contralateral right sensorimotor region, and vice versa. (2) A much more clear and significant attribution patterns are observed for Subject A with high BCI performance, reflecting localized MI features contributing to accurate BCI control. (3) As for cross-mode comparisons, Subject A exhibits consistent attribution patterns between the calibration and feedback stages whereas Subject B displays less consistent distributions. This explains why Subject B suffers from a severe performance degradation during online BCI control since discriminative brain patterns captured by models are subject to alteration as the mode changes.

D. Limitations and Future Work

There remain several limitations to be further explored in future work. While a remarkable improvement in performance using deep learning is observed over the initial sessions of MI-BCIs manipulation, longitudinal BCI training is also a necessity especially for BCI-inefficient populations [35]. As preliminarily demonstrated in this study, long-term training employing deep learning models might substantially improve the course of BCI skill acquisition and promote the utility of BCIs. Another limitation of this study is the use of structured and synchronized 2D center-out tasks. It is of great interest to explore the potential of deep learning-based BCIs to more sophisticated daily tasks [73] and more importantly, for potential end-users, e.g., stroke patients [3]. Third, while recent studies have leveraged attention mechanisms to cope with

subject and session variabilities for emotion recognition [74], attention modeling typically demands more training data and is prone to overfitting under limited EEG trials. It remains a question whether a particular efficient attention building block (e.g. transformer) would further benefit IFNet in a complementary sense for MI decoding. Last but not least, BCI preparation efficiency also plays an important role in eventual clinical applications. To boost preparation efficiency, further investigation on the minimal number of electrodes and calibration trials to meet expected feedback performance is warranted. In particular, transfer learning offers promising alternatives to deal with cross-subject and cross-session issues, effectively reducing the effort of BCI deployment required for online operation [75], [76].

VI. CONCLUSION

In conclusion, we implement deep learning models in online MI-BCIs and systematically validate the superiority of deep learning against prevailing machine learning approaches across sessions. Existing online MI-BCI studies predominantly adopt machine learning algorithms for BCI decoding and training. The study provides the first evidence that deep learning architectures designed by offline analysis indeed generalize well and achieve significantly improved online performance compared with existing benchmarks. Significant cross-session training effect is also observed by the deep learning model while not for the classical counterpart. Furthermore, we perform extensive and detailed behavioral and neurophysiological investigations of human-machine interaction underlying online BCI control. The results are pertinent to guide the future development of deep learning-based MI-BCIs.

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